Sentenced Differently: Project Steps

Phase 0: Setup (Week 1)

Goal: Get everyone oriented and the data pipeline running before any analysis starts.

Data acquisition

Download the U.S. Sentencing Commission individual offender datasets from the USSC website. Get the most recent 3 to 5 years (FY2020 to FY2024) in CSV format. Each file has one row per offender with variables for offense type, severity level, criminal history score, guideline range, actual sentence, and demographics.

Read the variable codebooks before writing any code. The variable names are not intuitive.

Repo structure

```
sentenced-differently/
  data/
    raw/              # USSC annual files, untouched
    processed/        # cleaned, merged output
  notebooks/
    01_eda.ipynb
    02_cleaning.ipynb
    03_regression.ipynb
    04_offense_breakdowns.ipynb
    05_visualizations.ipynb
  website/
    index.html
    about.html
    methodology.html
    assets/
      css/
      js/
      data/           # embedded JSON for charts and tool
  outputs/
    figures/
  README.md
  requirements.txt
```

Install: `pandas`, `numpy`, `statsmodels`, `scikit-learn`, `altair`, `scipy`

Key variables to locate in the codebook

Variable	What it measures
`MONCIRC`	Circuit of sentencing court
`OFFTYPE2`	Primary offense category
`GLMIN` / `GLMAX`	Guideline minimum and maximum (months)
`TOTPRISN`	Actual prison sentence imposed (months)
`CRIMHIST`	Criminal history category (I to VI)
`HISPORIG`	Hispanic origin
`NEWRACE`	Race code
`MONSEX`	Gender
`AGE`	Age at sentencing
`EDUCATN`	Education level
`CITIZEN`	Citizenship status
`ACCGDLN`	Whether sentence is within, above, or below guideline range

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Phase 1: Cleaning and EDA (Weeks 2 to 3)

Cleaning decisions to document explicitly

- Filter to cases where a prison sentence was imposed. Cases with probation only or zero months have a different data structure and should be excluded with a written justification.
- `TOTPRISN` includes life sentences coded as a large number. Cap at 360 months or log-transform and note the skew. Pick one approach and stick with it.
- Race and Hispanic origin are separate variables. Combine them into one: Non-Hispanic White, Non-Hispanic Black, Hispanic, Other. Document this construction in the methodology page.
- Drop cases with missing guideline range. A missing range usually means the case is atypical in ways that make comparison unreliable.

EDA outputs (saved as figures for the website)

- Distribution of sentence length overall and by year
- Sentence length by race/ethnicity and gender, unadjusted
- Distribution of offense types
- Criminal history distribution by demographic group
- Guideline departure rates by demographic group

The unadjusted gaps are the starting point. The whole analysis is about how much of those gaps survive after controls.

Phase 2: Regression Modeling (Weeks 3 to 5)

Model structure

Run models in stages. Each stage adds more controls. Show results at every stage so visitors can see how the demographic gap changes as legal factors are added.

Stage 1: Demographic variables only

```
log(sentence) ~ race + gender
```

Stage 2: Add offense type

```
log(sentence) ~ race + gender + offense_type
```

Stage 3: Add severity and criminal history

```
log(sentence) ~ race + gender + offense_type + guideline_min + crimhist
```

Stage 4: Full model

```
log(sentence) ~ race + gender + offense_type + guideline_min + crimhist + age +
education + citizenship + circuit
```

Log-transforming the outcome is standard here because sentence distributions are heavily right-skewed. Coefficients on the log scale translate to percent differences: a coefficient of 0.05 means that group received sentences about 5% longer on average, all else equal.

Stage 5: Interaction term

```
log(sentence) ~ race * gender + [all controls]
```

This is where the intersectional finding lives. Black men and women face different penalty structures, and some groups receive larger downward departures than others.

Export for the website

Once models are final, export coefficients and confidence intervals as JSON. The website will use these directly to power the comparison tool and the coefficient charts. No model should be re-run in the browser.

```python
import json

results = {
    "stage4": {
        "race_black": {"coef": 0.071, "ci_low": 0.058, "ci_high": 0.084},
        # etc.
    },
    "by_offense": { ... }
}

with open("website/assets/data/model_results.json", "w") as f:
    json.dump(results, f)
```

Robustness checks

- Run the same models on each year separately to check whether gaps are stable or changing
- Run on the two or three largest offense categories separately
- Check whether results change if you include or exclude cases with above- or below-guideline departures

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Phase 3: Offense Type Breakdowns (Week 5)

Subset to the five largest offense categories by case count. For each one, run the Stage 4 model and extract the race and gender coefficients. Build a coefficient plot showing the demographic penalty across offense types side by side.

Also compute departure rates separately. A departure is when a judge sentences outside the guideline range. Disparities can appear both in the sentence length and in who gets the benefit of a downward departure. These are related but distinct findings worth presenting separately.

Export all breakdown results to the same JSON file used by the website.

Phase 4: Website Build (Weeks 6 to 7)

The website has four pages. It is deployed via GitHub Pages (free, no backend needed). All data is embedded as JSON so there are no API calls or server dependencies.

Pages

`index.html` — Home

- Project title and one-paragraph plain-language summary
- Three or four KPI callouts: total cases analyzed, year range, headline racial gap, headline gender gap
- Navigation to the other pages
- A single striking chart (the unadjusted gap bar chart) above the fold

`dashboard.html` — Findings

Full scrolling dashboard with seven sections:

1. Unadjusted sentence gaps by race/ethnicity and gender
2. Regression results across model stages (coefficient plot showing gap shrinking as controls are added)
3. Intersectional results (race x gender interaction)
4. Offense type breakdown (coefficient plot across crime categories)
5. Departure rate analysis
6. Time trends (is the gap getting larger or smaller?)
7. Geographic breakdown by circuit

All charts built with Chart.js. Data embedded as JSON. No external data calls.

`tool.html` — Sentence Comparison Tool

This is the main interactive piece. A user sets a legal profile using dropdowns:

- Offense type
- Guideline minimum range
- Criminal history category
- Age group

- Education level
- Citizenship status

The tool then shows predicted sentence length for that exact profile across all demographic groups using the exported model coefficients. Display as a horizontal bar chart with confidence intervals shown as error bars. Add a plain-language summary below the chart: "For this profile, Black men are predicted to receive sentences X months longer than White men on average."

All computation happens in JavaScript using the stored coefficients. No model runs in the browser.

`methodology.html` — How We Did This

Written for a mixed audience. Two levels of detail:

- Plain-language explanation at the top (what the data is, what regression does, why we control for legal factors)
- Expandable technical section below for readers who want the full model specification, variable construction decisions, and links to published literature this builds on

Include a downloadable link to the executed Jupyter notebook and the cleaned dataset (if USSC terms allow redistribution, which they do).

Deployment

```bash
# From the repo root
git init
git add .
git commit -m "initial build"
gh repo create sentenced-differently --public
git push -u origin main

# In GitHub repo settings, enable Pages from main branch /website folder
# Site is live at: https://[your-org].github.io/sentenced-differently
```

If you want a custom domain, add a `CNAME` file to the `website/` folder with the domain name.

Optional: Vercel for faster deploys

Connect the repo to Vercel. Set the root directory to `website/`. Every push to `main` redeploys automatically. Vercel is also free for public projects.

Phase 5: Statistics Summary

What to report

Finding	Test
Unadjusted sentence gap by group	Kruskal-Wallis + pairwise Mann-Whitney with Bonferroni correction
Regression coefficients	OLS with robust standard errors, t-stats and p-values
Departure rate differences	Chi-square test
Model fit	R-squared at each stage; the jump from Stage 1 to Stage 3 shows how much of the raw gap is explained by legal factors

Effect size framing for the website

Convert everything to months in addition to percent differences. A 5% longer sentence at a mean of 50 months is 2.5 months. That framing lands with a general audience in a way that log coefficients do not. The tool and dashboard should always show both.

Team Roles

Role	Tasks
Data lead	Download, clean, and build the analytic dataset. Construct the combined race/ethnicity variable. Document every decision for the methodology page.
Modeling lead	Run all five regression stages. Produce coefficient tables. Run robustness checks. Export results to JSON.
Visualization lead	Build all Chart.js charts. Own the dashboard page HTML.
Tool and site lead	Build the comparison tool in JS. Assemble the four-page site. Handle GitHub Pages deployment.

You build the cleaning notebook and model skeleton first so teammates have working code to extend rather than starting from scratch.

Methodological Notes for the Presentation and Methodology Page

Acknowledge what you cannot control for

The USSC data does not include information on plea agreements, quality of legal representation, or prosecutorial charging decisions. These are upstream factors that also produce disparate outcomes. Your regression controls for what a judge sees at sentencing, not for what happened before sentencing. Be explicit about this scope on the methodology page.

Point to existing literature

The Sentencing Project, the Federal Sentencing Reporter, and the USSC itself publish research on this. Frame your contribution as updating the analysis with recent data and adding the intersectional interaction term, not as discovering something new.

The null result is still a result

If the demographic gap disappears after controlling for legal factors in a particular offense category, that is a finding worth reporting on the dashboard. It means the guidelines are doing their job in that domain. Do not hide null results.

Deliverables

1. Executed Jupyter notebook (all cells run, zero errors)
2. Deployed public website with dashboard and comparison tool
3. 18 to 20 slide presentation with speaker notes
4. Written project report (MD format)